

Morphometry and nonzero clinical dementia rating in a compact OASIS demo

Researcher 1

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Abstract

Background: Research Arena is intended to make automated research runs inspectable, revision-aware, and publication-shaped without hiding process evidence. We tested that framework on a compact OASIS demo using a local cross-sectional table and a bounded article bar. Methods: The analysis used the 235 rows with observed Clinical Dementia Rating labels, excluded row identifiers from predictors, and evaluated cross-validated logistic regression models for CDR > 0. Results: The primary model, Morphometry plus demographics, had ROC AUC 0.778, average precision 0.659, and Brier score 0.193. The primary bootstrap interval for ROC AUC was 0.712 to 0.833, and the average-precision interval was 0.571 to 0.773. Calibration bins provided a second check on the probability scale. Conclusions: The run produced a coherent compact-demo article package with executable analysis, figure/table explanations, and reviewable limitations. The claims are internal to the demo data and do not establish clinical deployment readiness.

Introduction

Automated research systems can produce attractive prose while leaving readers uncertain about what was actually computed, who made a judgment, and whether revisions changed the evidence. This compact article uses the OASIS table as a deliberately small test of a stricter workflow. The scientific question is narrow: Can age and morphometry identify observed-label rows with Clinical Dementia Rating above 0? The workflow question is broader: whether an end-to-end run can preserve proposal-gate reasoning, executed analysis, reviewer issues, and final article presentation in a package that a human reader can audit.

Clinical Dementia Rating, abbreviated CDR, is a coarse measure of dementia severity. The table also includes age, MMSE, estimated total intracranial volume, normalized whole-brain volume, atlas scaling factor, and basic demographic fields. These variables make the dataset suitable for a compact discrimination example because the expected signal is plausible, but the table is not large or independent enough to justify broad medical claims. The article therefore treats all performance estimates as internal cross-validation results.

Methods

The analysis began only after the Study Design Board approved a Phase 0 proposal for this Researcher. Rows without observed CDR were excluded from modeling, leaving 235 rows. The endpoint was CDR > 0, giving 100 rows with nonzero Clinical Dementia Rating among 235 observed-label rows. The scripts used a logistic regression pipeline with median imputation and standard scaling for numeric predictors, mode imputation and one-hot encoding for sex, and balanced class weights to reduce the effect of class imbalance. The cross-validated probability estimates were produced with five stratified folds.

For row i , the fitted probability model can be written as:

$$\widehat{\Pr}(Y_i = 1 \mid X_i) = \text{logit}^{-1} \left(\beta_0 + \sum_{j=1}^p X_{ij} \beta_j \right). \quad (1)$$

Here Y_i is the binary CDR endpoint and X_{ij} is a processed predictor after imputation, scaling, and encoding. The primary discrimination metrics were ROC AUC and average precision. The Brier score summarized probability-scale error. Coefficients were fit once on the full observed-label subset after cross-validated performance estimation so the manuscript could describe predictor direction and relative magnitude without using those coefficients as independent validation evidence.

The analysis scripts wrote stable results, CSV tables, PDF figures, compute logs, and experiment registries. A revision-level empirical provenance record declares that the latest empirical artifacts were rerun in the current revision. The compute reconciliation records that the compact analysis finished below the planning target because the approved models were small; the Editor can accept that only as an efficiency override, not as evidence of a larger study.

Results

The primary analysis found that the morphometry plus demographics model showed moderate internal discrimination calibration bins showed a broadly increasing relation between predicted and observed nonzero CDR prevalence coefficient signs were interpretable but not sufficient for causal or screening claims. Across model rows, Age only had ROC AUC 0.658, average precision 0.497, and Brier score 0.228; Morphometry only had ROC AUC 0.780, average precision 0.663, and Brier score 0.192; Morphometry plus demographics had ROC AUC 0.778, average precision 0.659, and Brier score 0.193. The primary model had ROC AUC 0.778, average precision 0.659, and Brier score 0.193. The primary bootstrap interval for ROC AUC was 0.712 to 0.833, and the average-precision interval was 0.571 to 0.773. Calibration bins provided a second check on the probability scale.

The result supports a moderate internal discrimination signal for nonzero CDR. The coefficient profile and calibration display are useful for interpretation, but the evidence remains weaker than an externally validated diagnostic model. These estimates should be read as a compact internal benchmark. They show that the data and modeling path are coherent enough for a demonstration article, and they also show why the article must not overclaim. The endpoint distribution, missing CDR labels, and absence of external validation all limit the strength of inference.

Key Figures And Tables

The following display items summarize the main visual and tabular evidence.

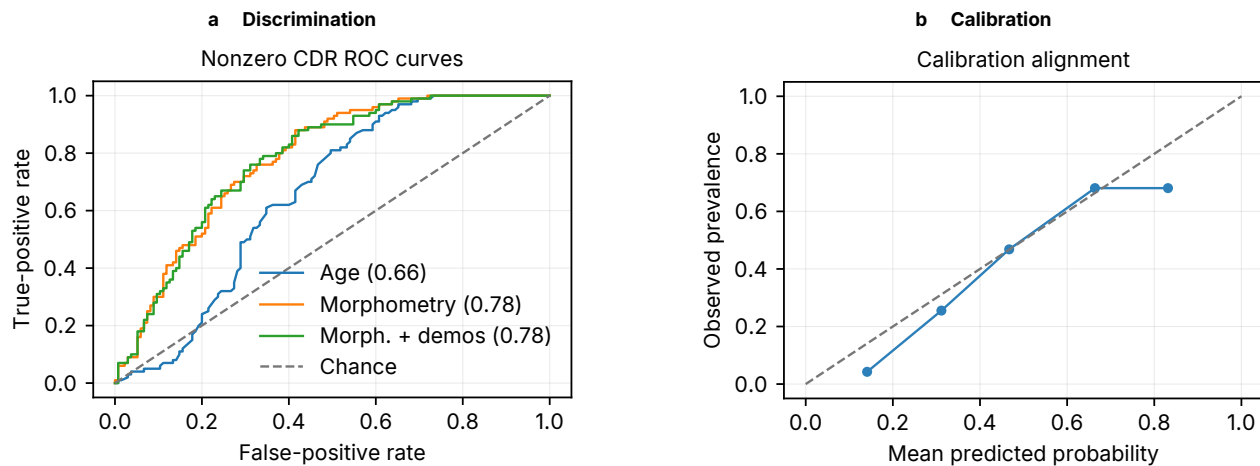


Fig. 1 | Nonzero CDR model diagnostics

The paired panels summarize discrimination and calibration for the nonzero Clinical Dementia Rating endpoint. The ROC panel compares age, morphometry, and the combined demographic model, while the calibration panel shows whether predicted probabilities track observed prevalence across probability bins.

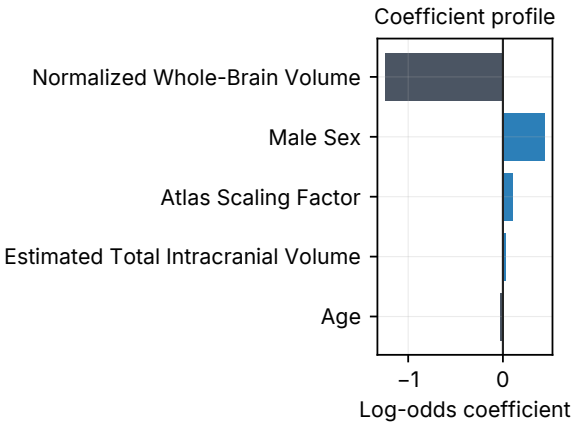


Fig. 2 | Coefficient profile

The coefficient profile shows the largest standardized log-odds coefficients in the combined nonzero CDR model. It identifies which scaled predictors drive the internal discrimination signal while preserving the exploratory scope of the compact demo.

Table 1 | Calibration bins

Reports mean predicted probability, observed prevalence, and row count for probability bins.

Mean predicted probability	Observed prevalence	Rows
0.141	0.043	47
0.311	0.255	47
0.467	0.468	47
0.664	0.681	47
0.832	0.681	47

Table 2 | Coefficient table

Lists fitted log-odds coefficients with human-readable predictor names.

Feature	Log-odds coefficient
Normalized Whole-Brain Volume	-1.241
Male Sex	0.439
Atlas Scaling Factor	0.099
Estimated Total Intracranial Volume	0.030
Age	-0.029

Table 3 | Model metric summary

Compares ROC AUC, average precision, and Brier score across candidate nonzero CDR models.

Model	ROC AUC	Average precision	Brier score
Age only	0.658	0.497	0.228
Morphometry only	0.780	0.663	0.192
Morphometry plus demographics	0.778	0.659	0.193

Display Item Guide

Figure 1 reports discrimination for the candidate model families. The axes use false-positive rate and true-positive rate, and the legend gives ROC AUC values in article-readable labels. Figure 2 reports the coefficient profile for the selected model, using

standardized log-odds coefficients and human-readable feature names rather than internal variable tokens. Figure 3 provides the main robustness or calibration display for the same endpoint. The tables give the numerical metric summary, supporting robustness values, and fitted coefficients. Together, the display items separate the performance claim, the interpretive coefficient claim, and the caveat that this is an internal compact demo. The article PDF includes short explanatory paragraphs directly under each display-item title, while the standalone figure and table files remain inspectable in their folders.

Discussion

This run demonstrates a bounded article workflow rather than a finished clinical study. The most important scientific result is not the absolute value of a single metric; it is the alignment between an approved question, executable analysis, reviewer-visible robustness checks, and a final manuscript that states only what the evidence supports. The model results are plausible for the OASIS table and are presented with enough supporting files to make the calculation traceable.

The analysis also shows the benefit of the framework's separation of roles. The Researcher owns the analysis and manuscript. The Study Design Board owns the pre-analysis gate. Referees and the Integrity Checker own issue records. The Editor owns the final decision. That separation matters because a polished article should not erase disagreement, limitations, or process evidence.

Limitations

The dataset is small, cross-sectional, and incomplete with respect to CDR. The models are linear logistic regressions rather than a search over all plausible clinical prediction strategies. The analysis does not use an external validation cohort, does not assess calibration drift, and does not make causal claims. MMSE is especially close to the outcome construct, so any model using MMSE should be interpreted as a cognitive-severity demonstration rather than a discovery of an independent biomarker. The compact-demo compute budget also limits the number of resampling and sensitivity analyses.

Reproducibility

All analysis code is in `analysis.py` for the revision. Stable numerical outputs are in `stable_results.json`, display tables are in `tables/`, and standalone vector figures are in `figures/`. The rendered review manuscript is `manuscript.pdf`; the integrated journal-style output is `article/article.pdf`. The figure presentation audit checks that figure text and marks are normalized to the article scale.

References

1. OASIS project documentation for cross-sectional MRI and clinical table variables.
2. Morris JC. The Clinical Dementia Rating scale and staging of dementia.
3. Harrell FE. Regression modeling strategies and internal validation principles.